

GABRIEL RODRIGUEZ

grodr358@fiu.edu ~ 7862386584 ~ www.linkedin.com/in/gabriel-rod ~ github.com/gabe0312 ~ <https://gabe0312.github.io/GabrielPortfolio/>

EDUCATION

Bachelor of Science in Computer Science

Graduated: April 2026

Florida International University; Miami, FL

- **GPA: 3.7 /4.0**
- **Certifications:**
 - Fundamentals of Robotics, Certification: Florida International University, 2026-04-30
 - Intermediate IOS Development, Certification: CodePath, 2025-12-17
 - Intro to Android Development, Certification: CodePath, 2025-08-07

TECHNICAL SKILLS

- **Embedded & Core Languages:** C++, C, Python, Go, Java, SQL, Flask
- **Hardware & Controllers:** ESP-32, Arduino, Raspberry PI, I2C, FreeRTOS
- **Tools & Technologies:** Git, GitHub, Azure DevOps, PlatformIO, VSCode, IntelliJ, React, Gradle, MongoDB, Flask, Gemini, Copilot, Codex
- **Web Development:** JavaScript, HTML, CSS, ReactJS, Kotlin, Swift

WORK EXPERIENCE

FIU Hardware Lab | Miami, FL

September 2024 - June 2025

Technical Lead Tutor

- Re-engineered lab sign-in workflow architecting an **HTML** interface, simplifying troubleshooting and significantly improving data accuracy.
- Demonstrated operational excellence by integrating **PHP** scripts to query student databases, reducing administrative errors by 30%.
- Fostered a culture of problem solving by mentoring 15 students weekly in object oriented design and data structures using **Java** and **C**.

Miami-Dade County IT | Miami, FL

May 2024 - May 2024

Software Intern

- Developed a redesigned employee intranet portal using **HTML**, **CSS**, and **JavaScript**, increasing employee access by 40%.
- Implemented **Azure DevOps** for version control and pull requests, within an agile environment, practicing CI/CD for seamless integration.
- Established a dynamic landing page with routing for newsletters and announcements, enhancing accessibility and navigation.

Zero Limit Learning | Miami, FL

June 2023 - August 2023

Stem Instructor

- Delivered STEM classes to K-12 students with lessons focused on coding robotic projects, achieving 95% lesson completion.
- Instructed students in technical design principles using **Tinkercad** and **Scratch**, translating concepts into accessible applications.
- Led teams in logic-based programming and sensor-based data collection, culminating in successful functioning demonstrations

PROJECTS

Panther Robotics Lunar Rover - Embedded Engineer | github.com/FIU-Lunabotics/FIU-Luna1.git

September 2025 - May 2026

- Engineered software-hardware integrations across a multidisciplinary team for rover system development, ensuring seamless operation.
- Deployed server logic on a **Raspberry Pi** to deserialize data to an **Arduino** board for precise low-level control of motors.
- Developed a network pipeline in **Go** to relay operator commands from a client to the rover, ensuring reliable communication.

WearYourPosture - Embedded Engineer | github.com/gabenogu/WearYourPosture.git

February 2026 - April 2026

- Dubbed "Most Impactful", "Most Unique" and "Best Overall" project by NextEra at the 2026 FIU **INIT Build** Demo Day.
- Designed firmware for an **ESP32** posture tracking device, interfacing with a **MPU6050** IMU via an **I2C multiplexer**.
- Wrote register level sensor communication code in **C++** to configure the IMU, ensuring precise real-time data acquisition.
- Applied embedded systems concepts and **FreeRTOS** task polling to build reliable, health-focused technological solutions.

Job Application Tracker - Full Stack Developer | github.com/Gabe0312/job-application-tracker.git

January 2026 - April 2026

- Architected and implemented a full stack job tracking application using Python and Flask, enhancing the job searching experience.
- Engineered a relational MySQL database with CRUD capabilities to maintain data integrity, alongside a custom job matching algorithm.
- Integrated AI development tools to accelerate feature prototyping and debugging, documenting usage protocols to ensure clean code.

BoxLang – Open-Source Developer | github.com/elsafranco/BoxLang.git

January 2026 - March 2026

- Contributed to open-source systems management by enhancing BoxLang's debugging runtime and object inspection capabilities.
- Optimized recursive data rendering across multiple classes, improving diagnosing while maintaining strict breadth limits.
- Authored automated tests to validate depth limits and structural output, ensuring environmental design and stability.

EXTRACURRICULAR INVOLVEMENT

Panther Robotics | Member : Attended hardware and system design workshops hosted by peers to build foundational knowledge in robotics.

INIT | Member: Participated in multi disciplinary technical development events and collaborative problem solving workshops to refine skills.

